

LAB EXPERIMENT RECORD

SET 11 – PRODUCT CATEGORY ANALYSIS

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
Name	HDN PRAHARSHINI
Register Number	192424085
Date	6 october 2026
Day	Tuesday

Aim

To analyze and visualize product category sales using funnel charts, pie charts, tables, and an interactive RStudio dashboard.

Dataset

Category	Sales (\$)
Electronics	50000
Clothing	35000
Appliances	40000

PROGRAM 1: Funnel Chart – Product Category Sales

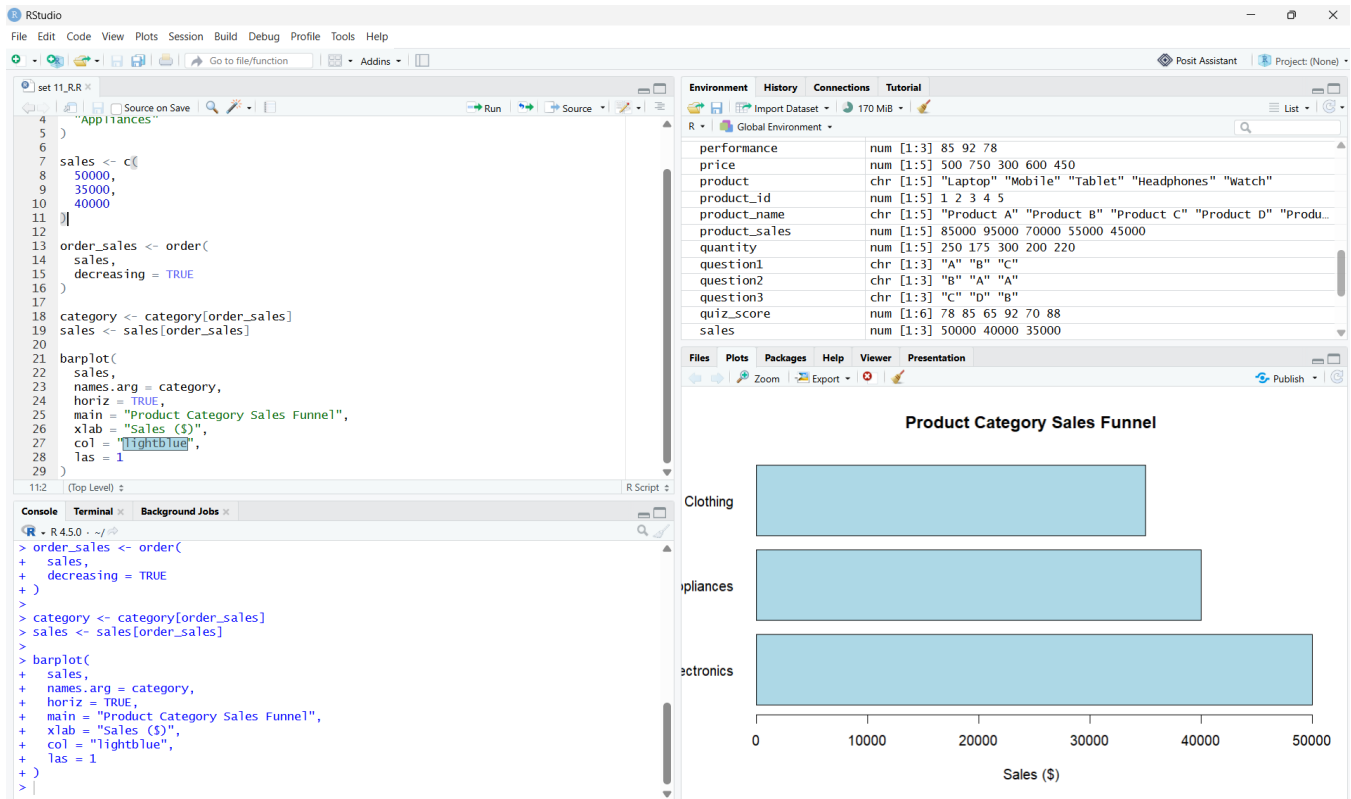
To create a funnel-style chart to compare sales across different product categories.

```
# category <- c(
  "Electronics",
  "Clothing",
  "Appliances")
sales <- c(
  50000,
  35000,
  40000)
order_sales <- order(
  sales,
  decreasing = TRUE
)
category <- category[order_sales]
sales <- sales[order_sales]
barplot(
  sales,
```

```

names.arg = category,
horiz = TRUE,
main = "Product Category Sales Funnel",
xlab = "Sales ($)",
col = "lightblue",
las = 1
)

```



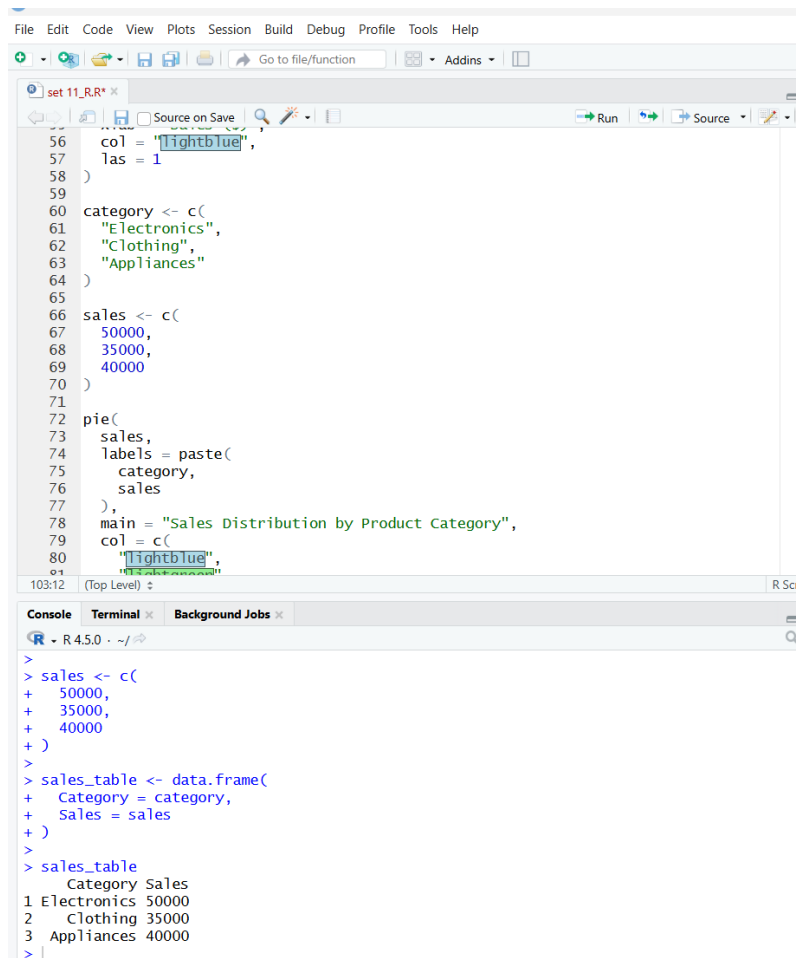
Output 1: Monthly Sales Line Chart

Observation: Electronics has the highest sales of **\$50,000**, followed by Appliances with **\$40,000**, while Clothing has the lowest sales of **\$35,000**..

PROGRAM 2: Sales table

To create a table displaying the sales information for each product category.

```
# category <- c(
  "Electronics",
  "Clothing",
  "Appliances")
sales <- c(
  50000,
  35000,
  40000)
sales_table <- data.frame(
  Category = category,
  Sales = sales)
sales_table
```



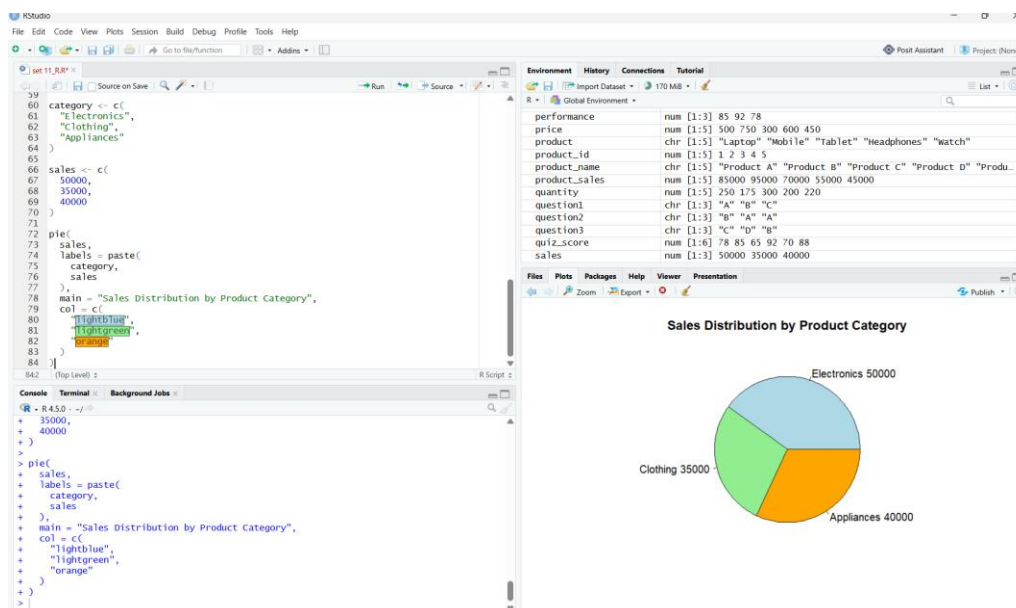
```
File Edit Code View Plots Session Build Debug Profile Tools Help
set 11_R.R* x
Source on Save
Run Source
56 col = "lightblue",
57 las = 1
58 )
59
60 category <- c(
61 "Electronics",
62 "Clothing",
63 "Appliances"
64 )
65
66 sales <- c(
67 50000,
68 35000,
69 40000
70 )
71
72 pie(
73 sales,
74 labels = paste(
75 category,
76 sales
77 ),
78 main = "Sales Distribution by Product Category",
79 col = c(
80 "lightblue",
81 "lightgreen",
82 "lightcoral"
83 )
84 )
103:12 (Top Level) R Sc
Console Terminal Background Jobs
>
> sales <- c(
+ 50000,
+ 35000,
+ 40000
+ )
>
> sales_table <- data.frame(
+ Category = category,
+ Sales = sales
+ )
>
> sales_table
  Category Sales
1 Electronics 50000
2 Clothing 35000
3 Appliances 40000
>
```

Observation: The table clearly displays the sales value for each product category

PROGRAM 3: Pie chart

To create a pie chart showing the distribution of sales across product categories.

```
# category <- c(
  "Electronics",
  "Clothing",
  "Appliances")
sales <- c(
  50000,
  35000,
  40000)
pie(
  sales,
  labels = paste(
    category,
    sales),
  main = "Sales Distribution by Product Category",
  col = c(
    "lightblue",
    "lightgreen",
    "orange"))
```



Observation

Electronics occupies the largest portion of the pie chart because it has the highest sales value.

PROGRAM 4: Interactive Sales Dashboard

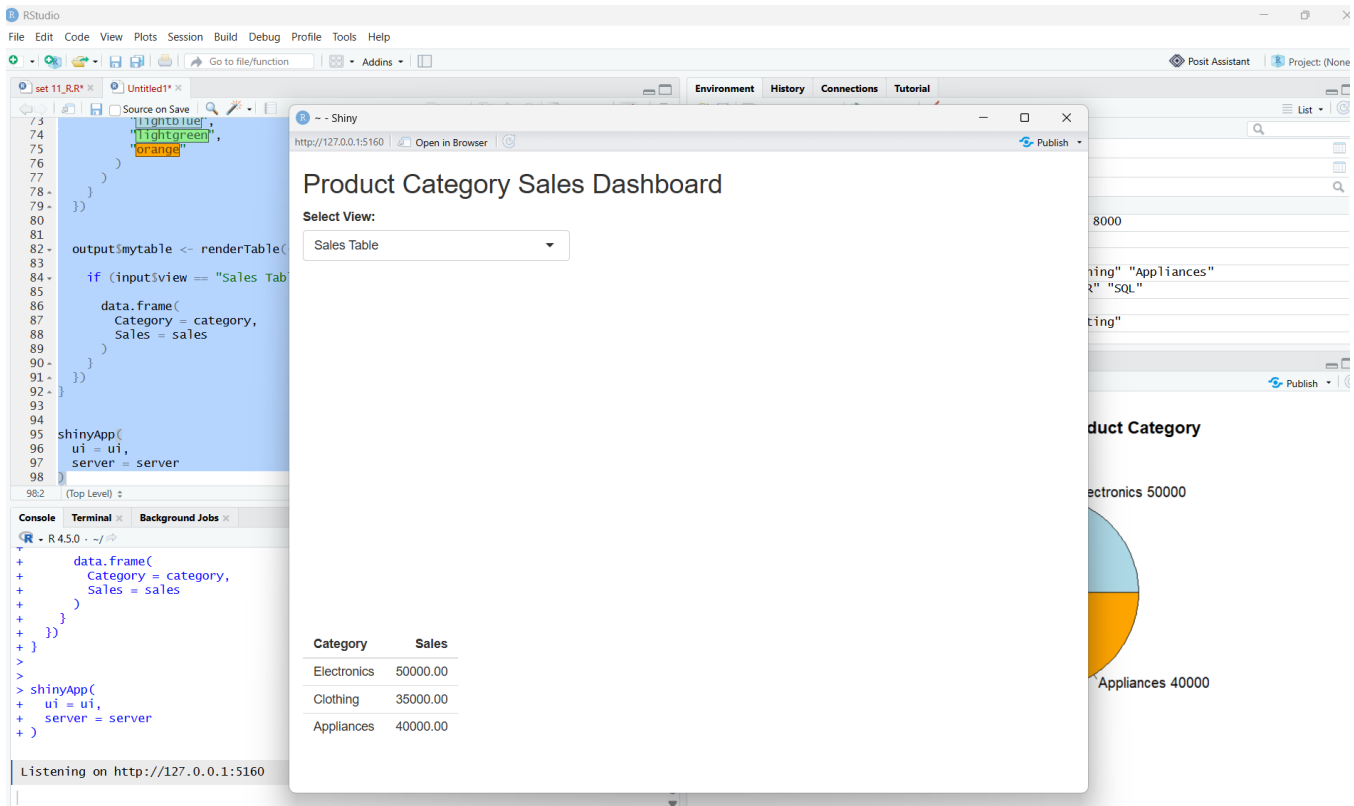
To develop an interactive dashboard combining the pie chart, funnel-style chart, and sales table.

```
library(shiny)
category <- c(
  "Electronics",
  "Clothing",
  "Appliances")
sales <- c(
  50000,
  35000,
  40000)
ui <- fluidPage(
  titlePanel(
    "Product Category Sales Dashboard"),
  selectInput(
    "view",
    "Select View:",
    choices = c(
      "Sales Funnel",
      "Sales Distribution",
      "Sales Table"
    )
  ),
  plotOutput("myplot"),
  tableOutput("mytable")
)
server <- function(input, output) {
  output$myplot <- renderPlot({
    if (input$view == "Sales Funnel") {
      order_sales <- order(
        sales,
        decreasing = TRUE
      )
      category_sorted <- category[order_sales]
      sales_sorted <- sales[order_sales]
```

```

barplot(
  sales_sorted,
  names.arg = category_sorted,
  horiz = TRUE,
  main = "Product Category Sales Funnel",
  xlab = "Sales ($)",
  col = "lightblue",
  las = 1
)
}
if (input$view == "Sales Distribution") {
  pie(
    sales,
    labels = paste(
      category,
      sales
    ),
    main = "Sales Distribution by Category",
    col = c(
      "lightblue",
      "lightgreen",
      "orange"    ))}
})
output$mytable <- renderTable({
  if (input$view == "Sales Table") {
    data.frame(
      Category = category,
      Sales = sales )
  }
})
}
shinyApp(
  ui = ui,
  server = server
)

```



Output 4: Interactive Monthly Sales Dashboard

Observation: The dashboard provides three views of product category sales: funnel-style comparison, sales distribution, and detailed sales table.

Procedure:

- Create vectors for product categories and sales.
- Sort the sales values and use a horizontal bar chart as a funnel-style visualization.
- Use `data.frame()` to create the sales table.
- Use `pie()` to display the distribution of sales.
- Use Shiny to combine the funnel-style chart, pie chart, and table.
- Save each question as a separate R script.

Result:

The product category sales data was successfully analyzed using a funnel-style chart, pie chart, and table. An interactive dashboard was also developed in RStudio to explore the product category sales